

Exact Fractional Hamiltonicity on the Branch $J_{m,1}$

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Abstract

For the graph $J_{m,\ell} = K_\ell \vee H_m$, where H_m is obtained by subdividing every edge ab_i of a complete graph on $\{a, b_1, \dots, b_m\}$ once, we determine the complete cut linear program on the infinite branch $\ell = 1$. For every integer $m \geq 2$, its optimum is

$$\text{OPT}_{m,1} = \max\{2m + 2, 5m/2\}.$$

It follows that $J_{m,1}$ is fractionally Hamiltonian precisely for $2 \leq m \leq 4$. In particular, $J_{5,1}$ is an exact boundary example one integer below the previously known sufficient negative range. The original classification for arbitrary (m, ℓ) remains unresolved.

1 Statement and context

All graphs in this note are finite, simple, and undirected. Let $B = \{b_1, \dots, b_m\}$ and $X = \{x_1, \dots, x_m\}$. The graph H_m has vertex set $\{a\} \cup B \cup X$ and edge set

$$\binom{B}{2} \cup \{ax_i, x_i b_i : i \in [m]\}.$$

Thus H_m is obtained from the complete graph on $\{a, b_1, \dots, b_m\}$ by subdividing precisely the edges ab_i . Define $J_{m,\ell} = K_\ell \vee H_m$. In the branch considered here, write u for the sole vertex of the joined K_1 ; hence $J_{m,1}$ has order $2m + 2$.

For a graph $G = (V, E)$ and a nonempty proper set $S \subsetneq V$, let $\delta_G(S)$ be the set of edges with exactly one endpoint in S . Consider the complete cut program

$$\begin{aligned} & \text{minimize} && \sum_{e \in E} x_e, \\ & \text{subject to} && \sum_{e \in \delta_G(S)} x_e \geq 2 \quad (\emptyset \neq S \subsetneq V), \\ & && x_e \geq 0 \quad (e \in E). \end{aligned} \tag{P}$$

Its optimum for $J_{m,\ell}$ is denoted by $\text{OPT}_{m,\ell}$. A graph of order $n \geq 3$ is fractionally Hamiltonian if and only if the optimum of (P) is n ; this cut-LP formulation is recalled by Wang [1, Lemma 7].

Wang introduced the same family $J_{m,\ell}$ in this setting, proved $\tau(J_{m,\ell}) = 1 + \ell/m$, and proved that it is not resistance nonnegative when $m \geq 2\ell + 4$ [1, Propositions 15–16]. Wang also proved that fractional Hamiltonicity implies resistance positivity [1, Theorem 4]; since resistance positivity implies resistance nonnegativity, these statements give a sufficient non-fractional-Hamiltonicity range beginning at $m = 6$ when $\ell = 1$. The result below identifies the exact threshold on this

branch, including the omitted value $m = 5$. It is a partial result: no assertion is made here about the full two-parameter classification or about absolute priority beyond the literature comparison just stated.

Theorem 1. *For every integer $m \geq 2$,*

$$\text{OPT}_{m,1} = \max \left\{ 2m + 2, \frac{5m}{2} \right\} = \begin{cases} 2m + 2, & m = 2, 3, \\ 5m/2, & m \geq 4. \end{cases}$$

Consequently, $J_{m,1}$ is fractionally Hamiltonian if and only if $2 \leq m \leq 4$. In particular, $\text{OPT}_{5,1} = 25/2 > 12$ and $\text{OPT}_{6,1} = 15 > 14$.

2 Lower bounds

Summing the singleton-cut constraints of (P) counts every edge twice. On a graph of order n this gives

$$2 \sum_{e \in E} x_e = \sum_{v \in V} x(\delta_G(\{v\})) \geq 2n.$$

For $J_{m,1}$, therefore, $\text{OPT}_{m,1} \geq 2m + 2$.

The dual of (P) assigns a nonnegative weight y_S to each distinct nontrivial cut and has objective and constraints

$$\text{maximize } 2 \sum_S y_S, \quad \sum_{S: e \in \delta(S)} y_S \leq 1 \quad (e \in E). \quad (\text{D})$$

For each $i \in [m]$, set

$$y_{\{b_i\}} = \frac{1}{4}, \quad y_{\{x_i\}} = \frac{3}{4}, \quad y_{\{b_i, x_i\}} = \frac{1}{4},$$

and set all other dual variables to zero. The resulting dual edge loads are 1 on edges of types $b_i b_j$, $a x_i$, $b_i x_i$, and $u x_i$; they are $1/2$ on $u b_i$ and 0 on $u a$. Thus the vector is dual feasible, and its objective is

$$2m \left(\frac{1}{4} + \frac{3}{4} + \frac{1}{4} \right) = \frac{5m}{2}.$$

Weak duality now yields

$$\text{OPT}_{m,1} \geq \max\{2m + 2, 5m/2\}. \quad (1)$$

3 A matching all-cut certificate

Suppose $m \geq 4$. Give the edges of $J_{m,1}$ the weights

$$x_{b_i b_j} = \frac{1}{m-1}, \quad x_{a x_i} = x_{u x_i} = \frac{1}{2}, \quad x_{b_i x_i} = 1, \quad (2)$$

and give $u a$ and all $u b_i$ weight zero. The total weight is

$$\binom{m}{2} \frac{1}{m-1} + m \left(\frac{1}{2} + 1 + \frac{1}{2} \right) = \frac{5m}{2}.$$

It remains to verify every cut. For a shore S , use $A, U, B_i, X_i \in \{0, 1\}$ for the membership indicators of a, u, b_i, x_i , respectively, and put $k = \sum_i B_i$. Under (2), the exact weight of its cut is

$$C(S) = \frac{k(m-k)}{m-1} + \sum_i |B_i - X_i| + \frac{1}{2} \sum_i |X_i - A| + \frac{1}{2} \sum_i |X_i - U|. \quad (3)$$

If $A \neq U$, the last two sums in (3) contribute exactly $m/2$, and hence $C(S) \geq m/2 \geq 2$.

If $A = U$, replacing S by its complement when necessary permits us to assume $A = U = 0$. If $k = 0$, nontriviality forces $X_i = 1$ for some i , and that index contributes $1 + 1/2 + 1/2 = 2$ to (3). If $k \geq 1$, each index with $B_i = 1$ contributes exactly 1 to the last three sums, so

$$C(S) \geq k + \frac{k(m-k)}{m-1} \geq 2.$$

For $k = 1$ the final expression equals 2; for $k \geq 2$ the inequality is immediate, including $k = m$. Thus (2) is feasible for every nontrivial cut. Its objective and (1) prove the asserted optimum for all $m \geq 4$.

For $m = 2$, the vertex sequence

$$a, x_1, b_1, b_2, u, x_2, a$$

is a Hamiltonian cycle. For $m = 3$, one is

$$a, x_2, b_2, u, x_1, b_1, b_3, x_3, a.$$

Their incidence vectors are feasible for (P) with objectives 6 and 8, respectively. The singleton lower bound proves optimality in both cases. This completes the proof of Theorem 1. Since the order is $2m + 2$, the cut-LP equivalence gives fractional Hamiltonicity exactly for $m = 2, 3, 4$.

4 Exact computational check and limitations

As a check on the displayed certificates, a separate exact-arithmetic script expanded orbit-constant edge weights and enumerated one shore from every complementary pair for $m = 2, 3, 4, 5, 6$. It checked respectively 31, 127, 511, 2047, and 8191 nontrivial cuts, as well as every dual edge constraint. All arithmetic used rational numbers; in each instance the minimum primal cut weight was 2, the maximum dual edge load was 1, and the primal and dual objectives agreed. These finite computations audit the formulas but are not used to extrapolate the theorem: the quantified $m \geq 4$ conclusion rests on (3). This separation between a symbolic all-parameter argument and proof-carrying finite checks follows the same certificate-first methodological principle used in exact finite combinatorial models by Zeng et al. [2]; that work is not prior literature on fractional Hamiltonicity.

The theorem settles only the definition-closed branch $\ell = 1$. In particular, it neither supplies the value of $\text{OPT}_{m,\ell}$ for $\ell > 1$ nor determines the full fractional-Hamiltonicity threshold in the original two-parameter problem.

Computational and AI assistance. OpenAI Codex assisted with exact rational certificate checking, development of the symbolic cut argument, and manuscript preparation. The accompanying verification script uses Python’s `fractions.Fraction`; its serialized certificate inputs and captured output record the finite checks described above. All claims in the theorem are supported by the explicit symbolic proof rather than by numerical fitting or finite extrapolation.

References

- [1] Zhiyu Wang. Toughness bounds for fractional hamiltonicity and resistance positivity, 2026. arXiv:2609.03412v1, submitted 3 September 2026.

- [2] Zijian Zeng, Houde Liu, Kuru Ratnavelu, and Ong Seng Huat. Exact finite certificates for the four-dimensional ternary borsuk problem and five-dimensional ternary kissing codes. *Preprints*, August 2026.